



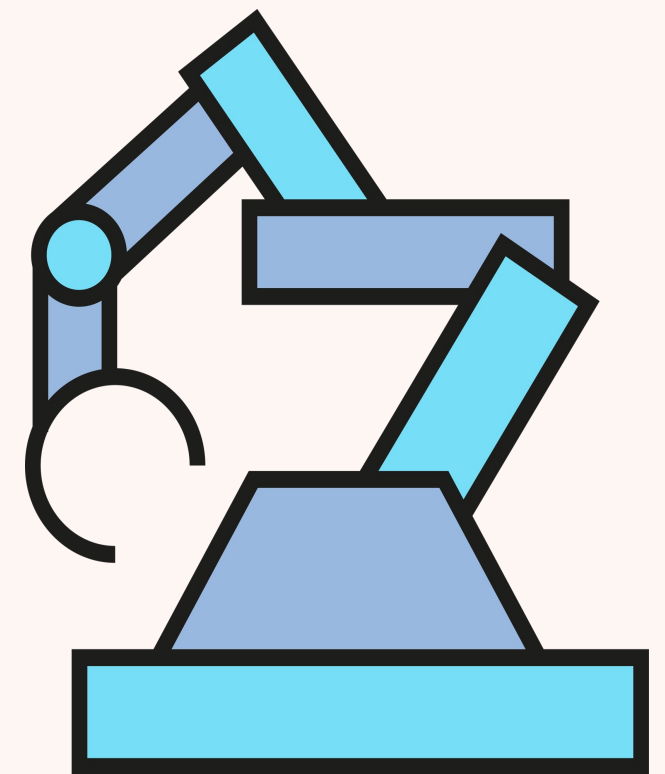
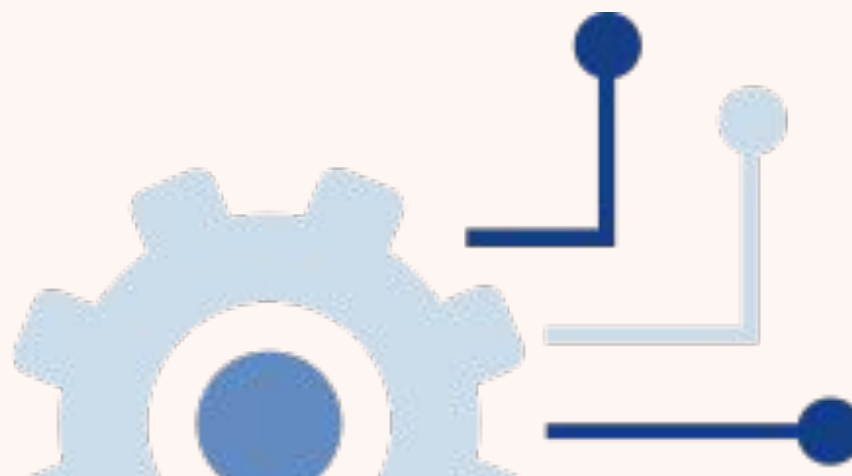
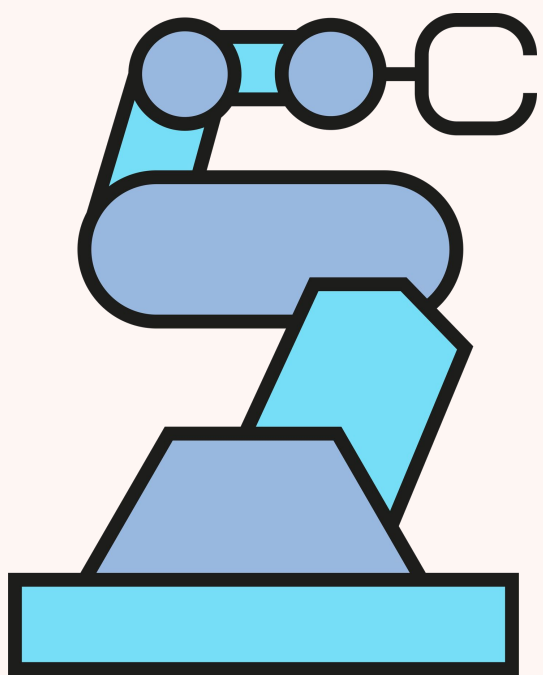
Training Bimanual Handoff using UMI

Stanford UMI pipeline adapted for RX150 arms

**SONOMA
STATE
UNIVERSITY**

INTELLIGENT
SYSTEMS LAB

Nicholas Frangione • Kai Nunnemaker
In collaboration with Dr. Sudhir Shrestha & Antonio Lozoida Cigala



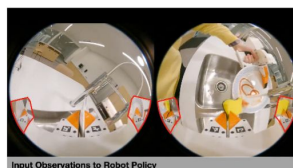
What Stanford UMI Gives Us



Universal Manipulation Interface

In-The-Wild Robot Teaching Without In-The-Wild Robots

Human Demonstration with UMI



Fully Autonomous Policy Rollout



UMI is a demonstration-to-policy framework. A person uses handheld grippers with cameras, the system reconstructs motion from video, and a visuomotor model learns how to imitate the task on a robot.

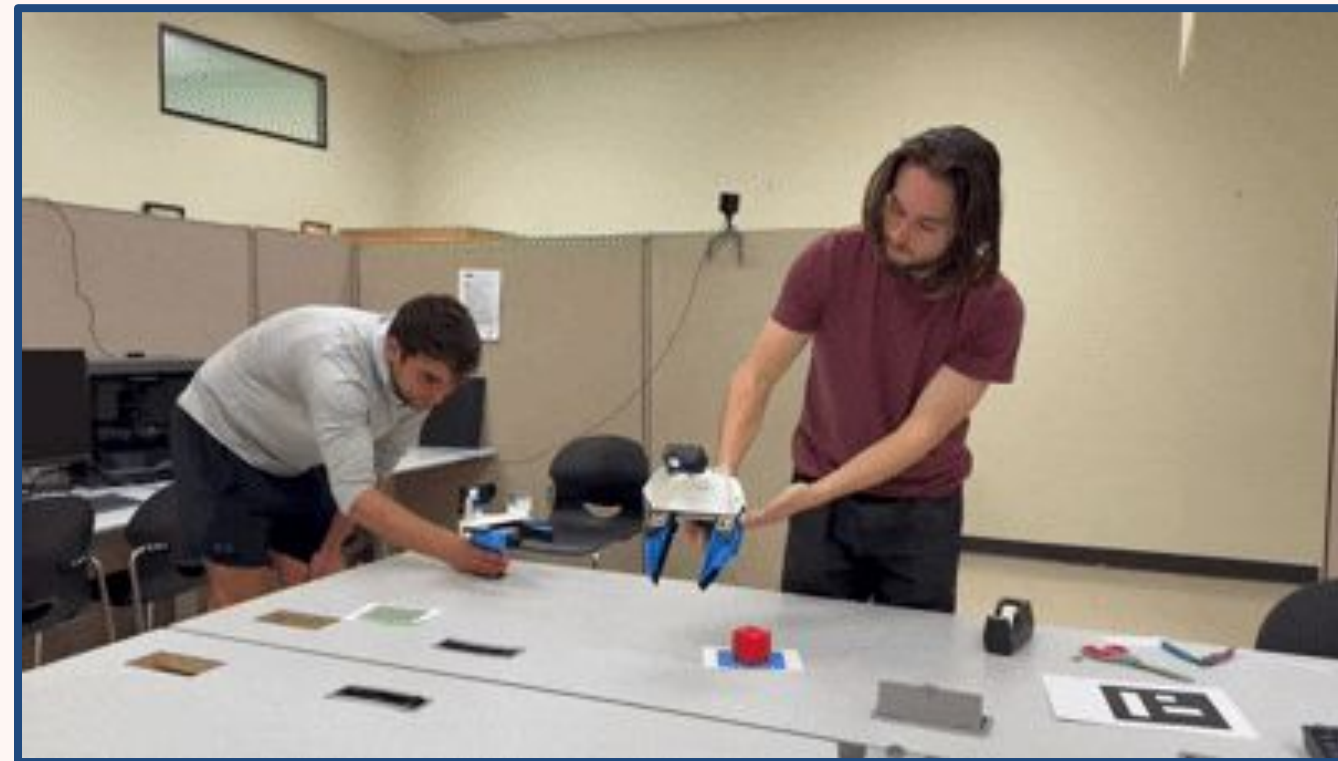
**For our project:
train a two-arm handoff
task**

**Key difference:
RX150 has 5 motion DOF**

So the final product needed more than model training. It needed a hardware translation layer that turns predicted UMI-style actions into movements our arms can actually execute.

The Task we Chose!

Bimanual Handoff



Main Project Pipeline

Human Demo to Robot Action Sequence

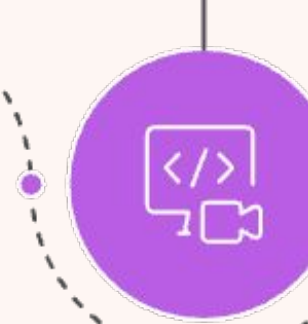
Build UMI Gripper Hardware

Constructing the gripper with necessary components



Record Human Demonstrations

Capturing human actions for training



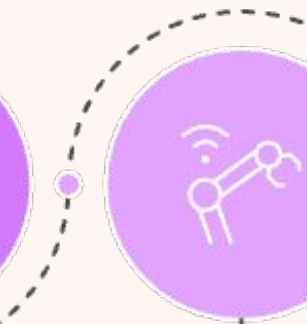
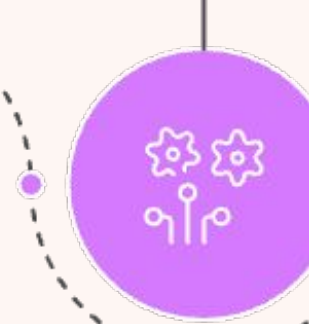
Process Video with SLAM

Estimating camera and gripper motion



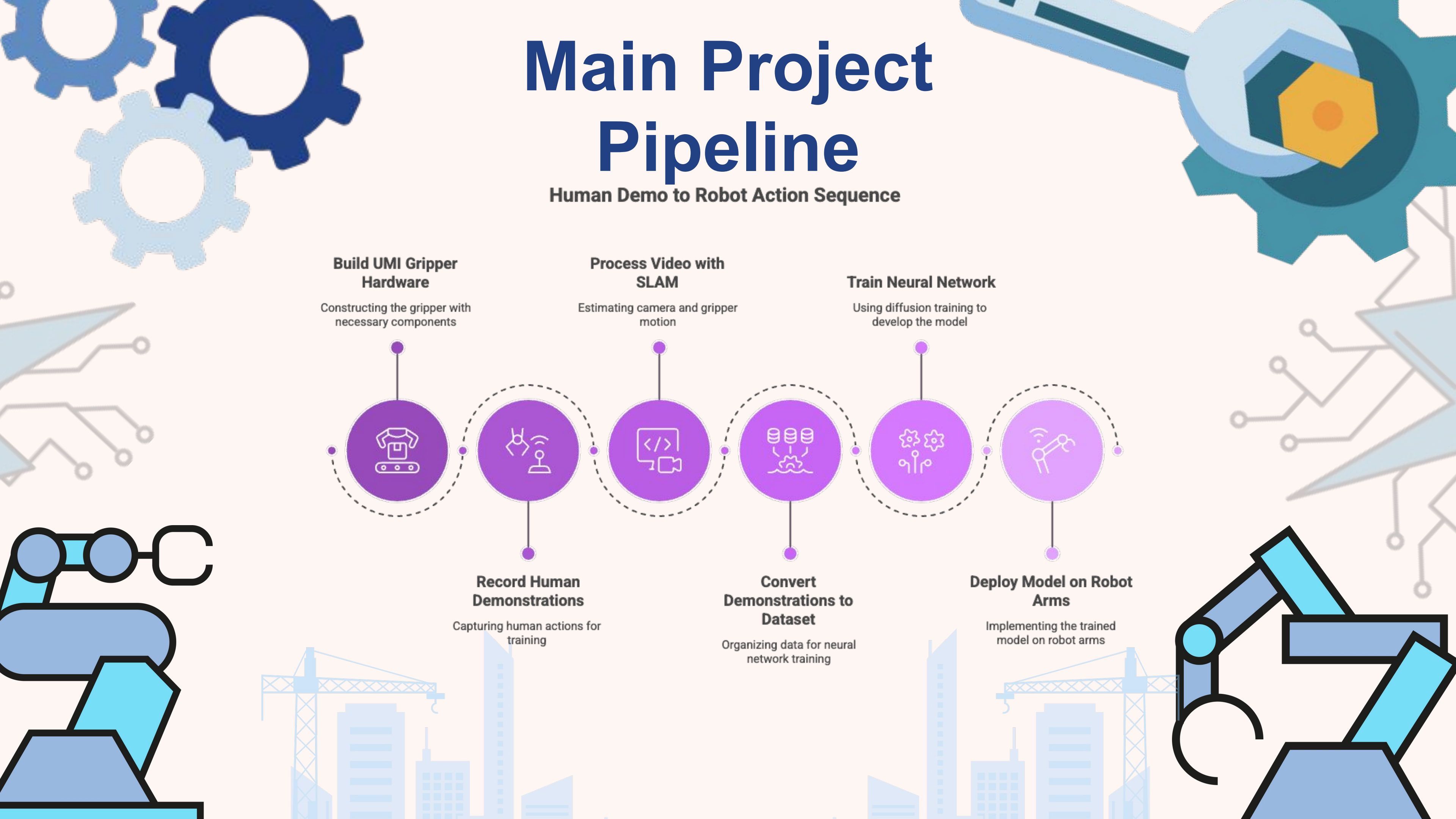
Train Neural Network

Using diffusion training to develop the model

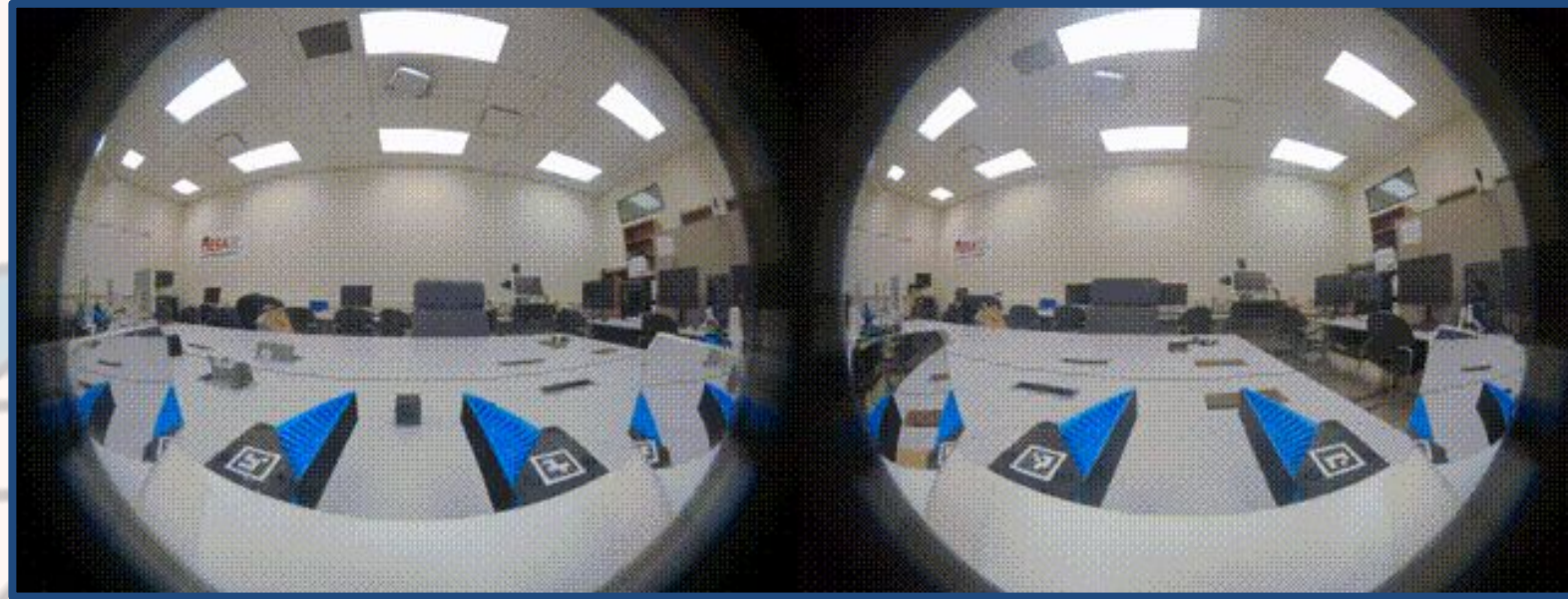


Deploy Model on Robot Arms

Implementing the trained model on robot arms



Visual Input: What the Cameras and Grippers Record



Paired GoPro views from the workspace

Camera frames
visual scene around each hand

IMU / timing
motion cues and synchronization

Marker detections
pose anchors for gripper tracking

Task context
block, table zones, and handoff geometry

Output of this stage: synchronized, paired observations that can be turned into training examples.

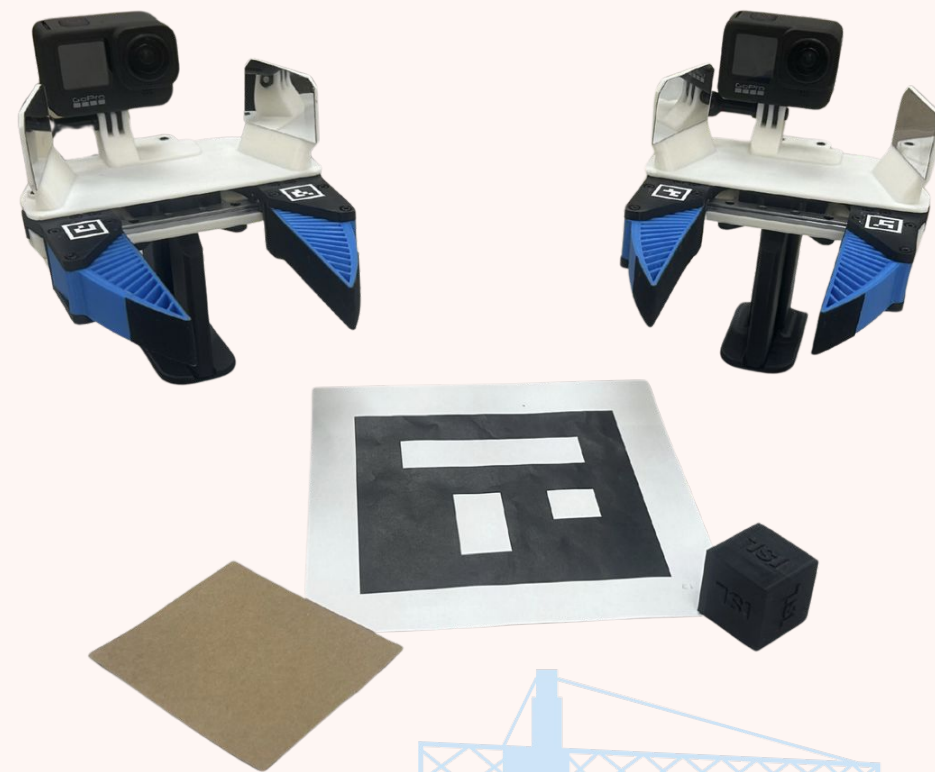
Data Collection

Goal:

Collect paired demonstrations of a block being moved from one arm's workspace to the other.

What Happened:

We collected roughly 300 videos per arm across a total of 5 rounds.



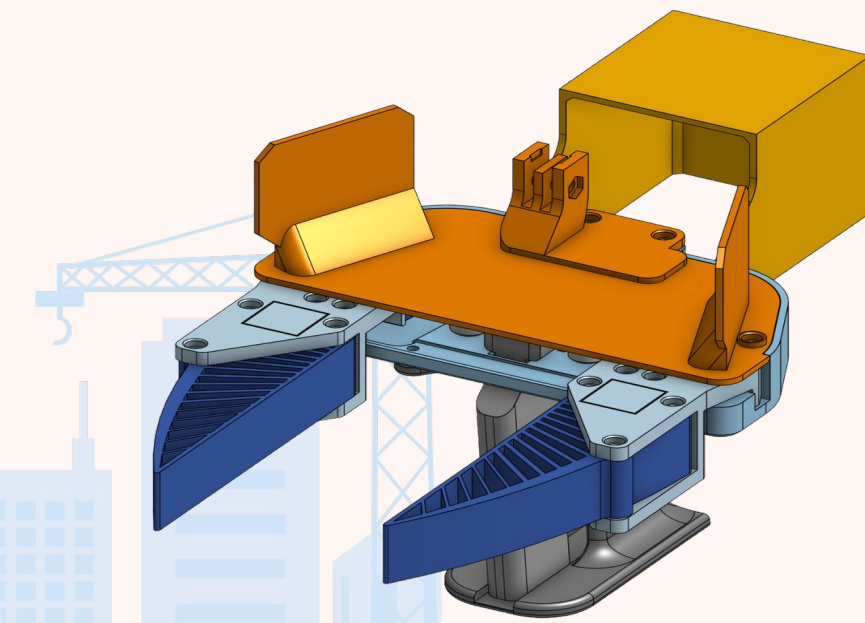
Two UMI grippers + ArUco/QR targets + task object

Main issues:

- GoPro time sync problems
- Two separate mapping videos caused SLAM issues
- Covered mirrors on the UMI grippers
- Undersized or duplicated QR codes

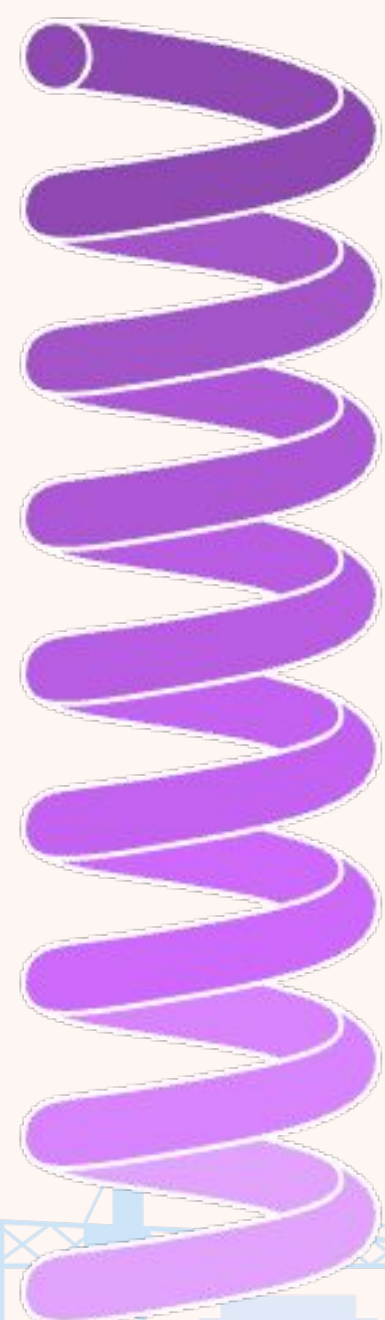
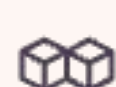
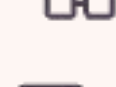
Final fixes:

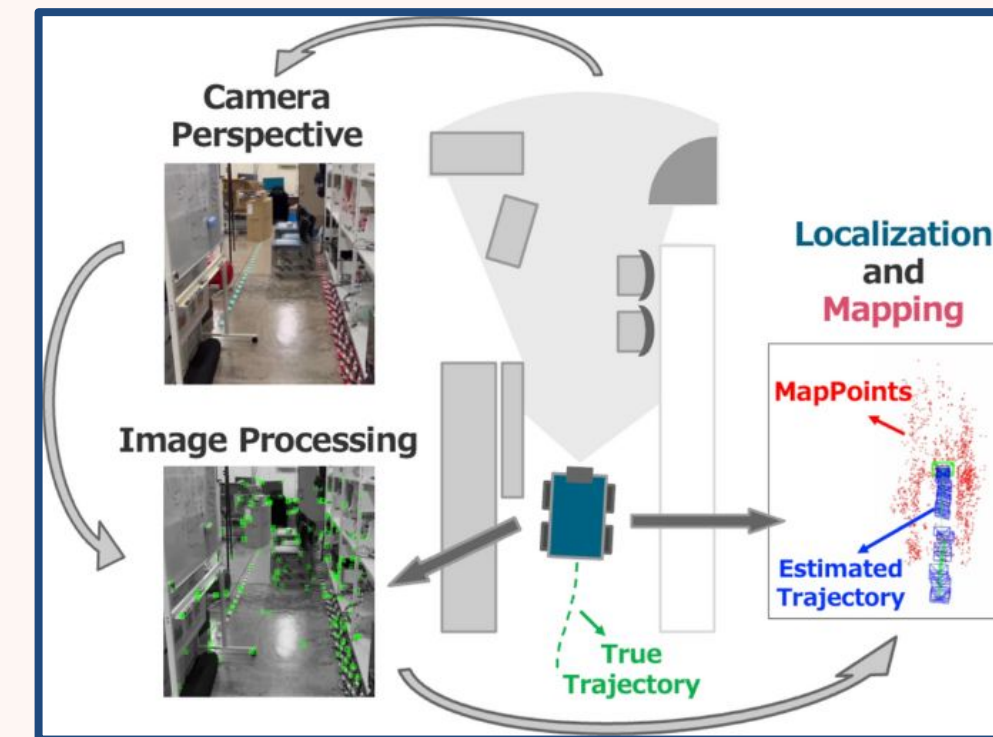
- Resynced cameras every 10 demos
- Used one longer mapping video
- Replaced QR codes so each arm could be detected separately



SLAM pipeline

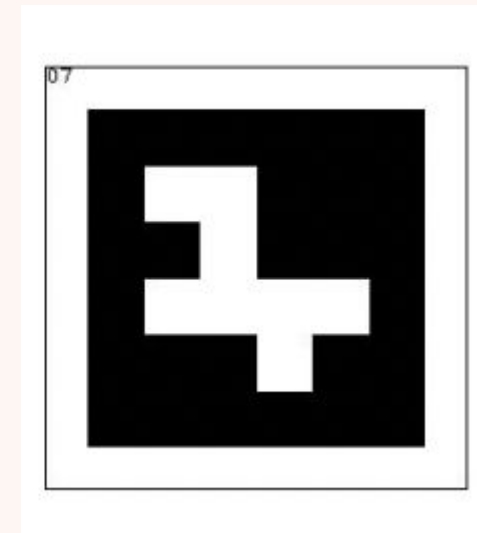
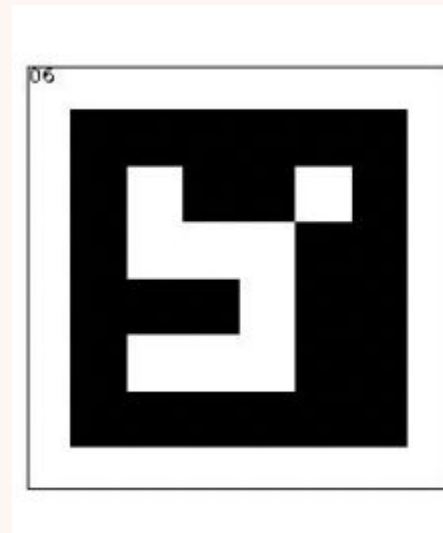
SLAM Data Processing Sequence

- 
-  Process Videos
 -  Extract GoPro Imu
 -  Create Mapping
 -  Batch Slam
 -  Detect Aruco
 -  Run Calibrations
 -  Dataset Plan
 -  Generate Replay Buffer



SLAM Pipeline Results

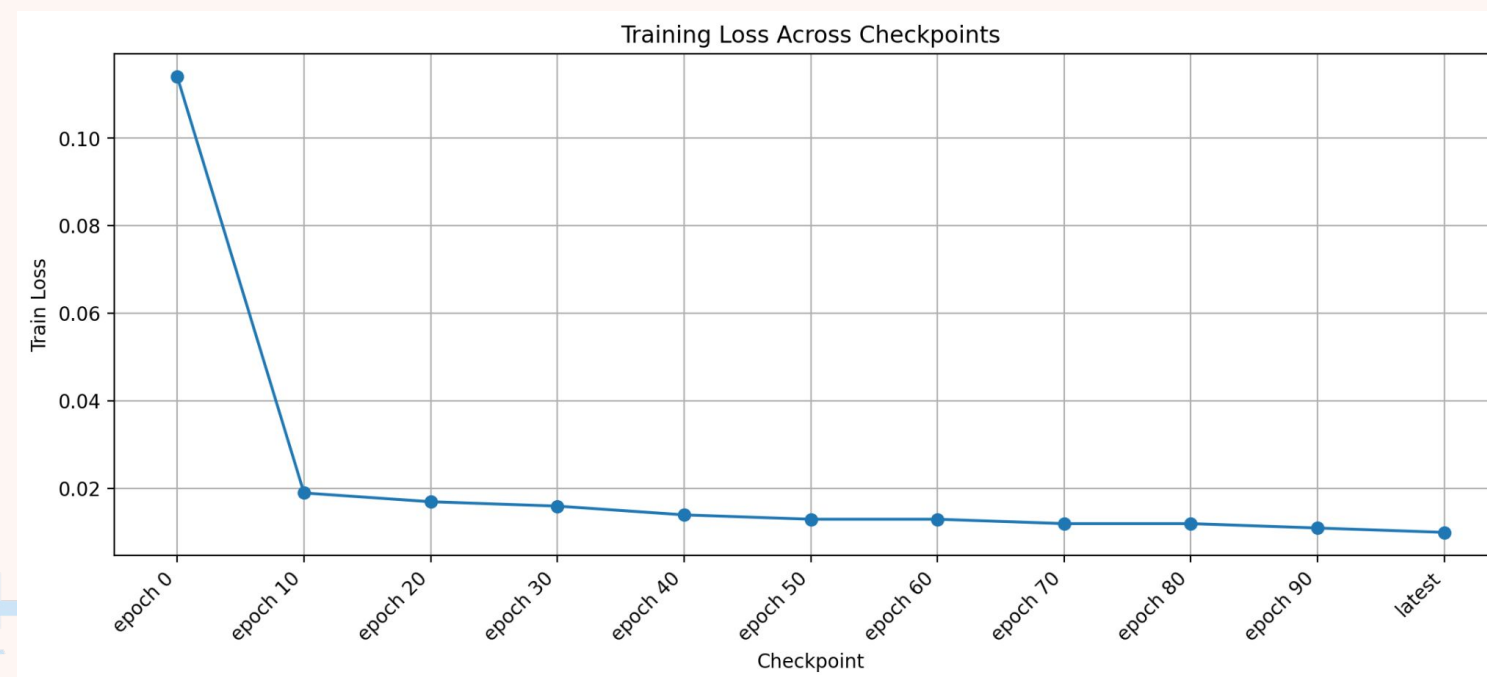
- The SLAM pipeline processed the GoPro footage into usable robot-training data.
- Final result:
 - 101 newest demos processed
 - 55 paired demos successfully generated
- Solved:
 - QR code detection issues
 - Camera time misconfiguration
 - More reliable paired demonstrations
- Still limited by:
 - Occasional inconsistent camera trajectory generation
 - Mapping video quality and workspace setup



Training

NRP NATIONAL RESEARCH PLATFORM

- We uploaded the successful paired demos to the National Research Platform and trained a UMI diffusion policy model. based on the most recent set of 55 paired demos

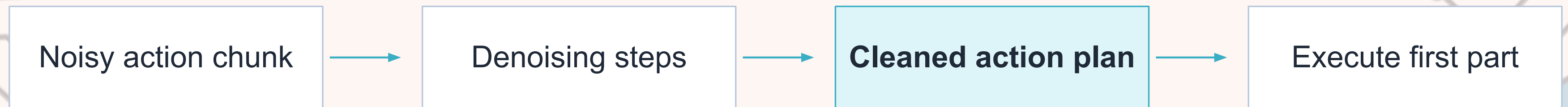


Training Process

- Each training sample contains two camera views, robot/gripper states, relative bimanual pose features, and a future action chunk.
- CLIP-pretrained Vision Transformer encoded camera images, combined with low-dimensional robot state inputs
- By adding noise to ground-truth action sequences, it trains the network to denoise/predict the correct robot actions.
- Training loss decreased from 0.114 to 0.010 over 100 epochs (44,900 Sample/Epoch), showing the model was learning

What the Diffusion Policy Is Doing

Instead of predicting one single next move, the policy predicts a short future sequence of actions. During training, it learns to remove noise from correct action sequences.



For robotics, this is useful because there may be multiple valid ways to complete a handoff. The model can represent several plausible action patterns, then commit to one during rollout.

What Goes Into the Model and What Comes Out

Model Inputs

Two camera views

Robot / gripper states

Relative bimanual pose features

Timing-aligned samples

Diffusion Policy

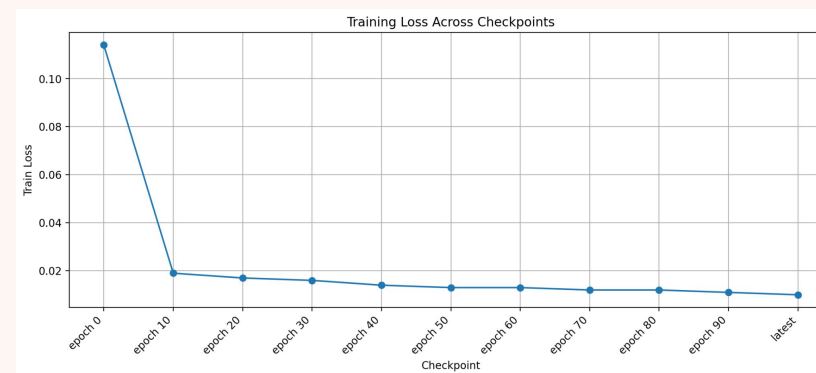
Model Outputs

Future action chunk

Target hand / gripper motion

Relative motion commands

Receding-horizon updates



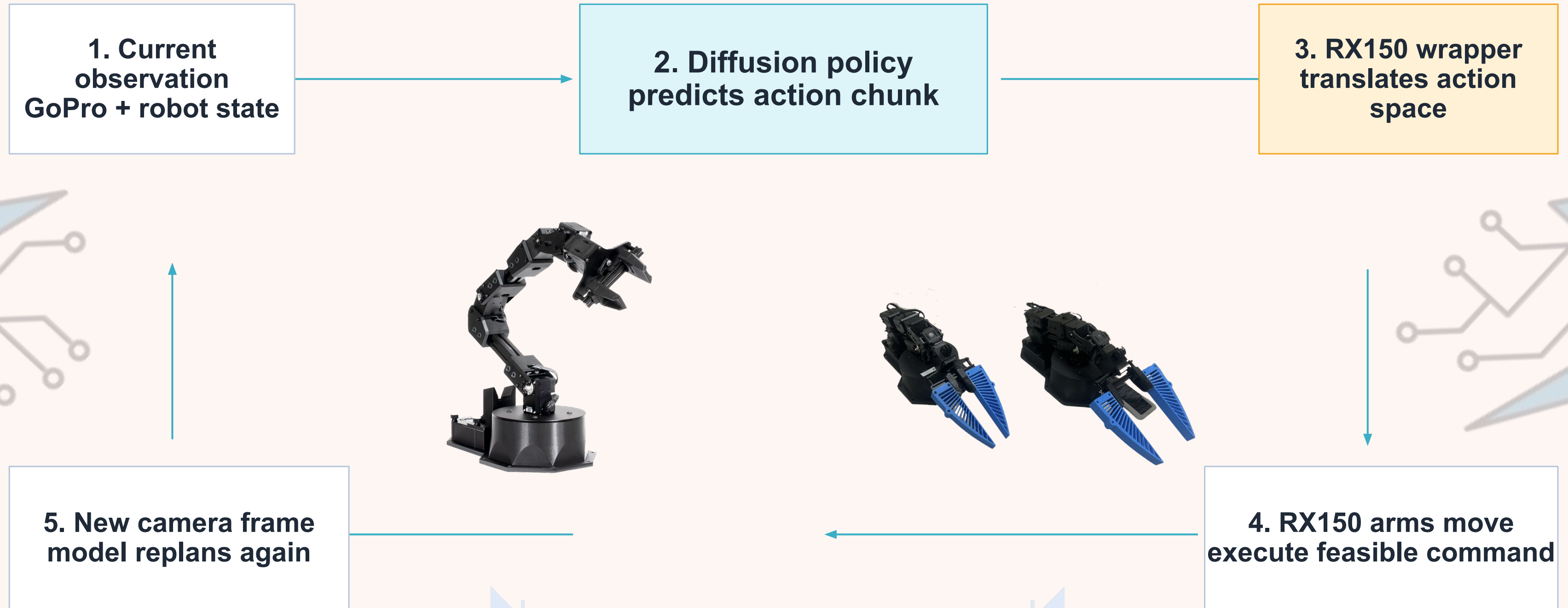
Training loss dropping showed the model was fitting the demonstration data

Training result: loss decreased from 0.114 to 0.010 over 100 epochs.

Quantitative Analysis

- Quantitative Evaluation
 - We evaluated the trained policy model by comparing predicted actions against ground-truth actions from recordings
- Main result:
 - Training dataset MAE: 0.00132
 - Testing dataset MAE: 0.00455
 - Training dataset L2 error: 0.00805
 - Testing dataset L2 error: 0.04938
- Takeaway: The model performs much better on training data than testing data (which contained incorrect qr codes), suggesting possible overfitting and low ability to generalize.

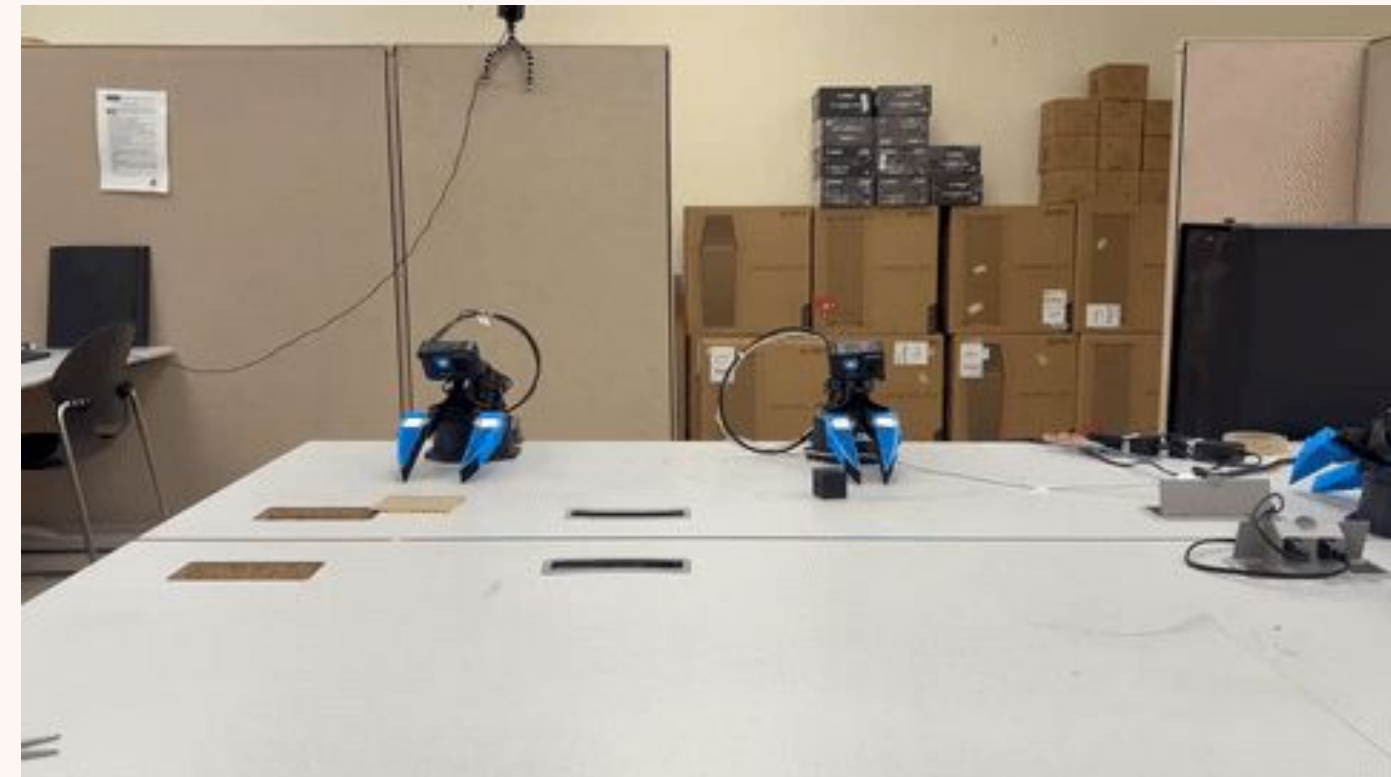
Closed Loop: Predictions Become Arm Motion



The policy does not “finish the whole task at once”; it repeatedly updates as observations change.

Physical Implementation

- Limitations with Physical Implementation:
 - Robot Arm (RX-150) different than the Arm utilized in Stanford Experiment
 - Lack of Mirrors on Deployed Arms
 - Not a way to secure training mirrors on our arms
 - Motor Strength Capacity Limitations
 - Limiting Movement
 - Camera Mount Position
 - Different mount angle
 - Cheaper HDMI->USB Dongle
 - Video feed lag





Welcome to QnA Time

Data Collection

- Pick up the block from the blue square and place on the green square
- Collect roughly 50 videos an additional 50 if needed
- Begin the SLAM and training process

- **The Goal**

What Happened -

- 4 Different Rounds of Collection
- roughly 300 videos (per Arm)
- And Many Many hours spent figuring out why SLAM was not succeeding

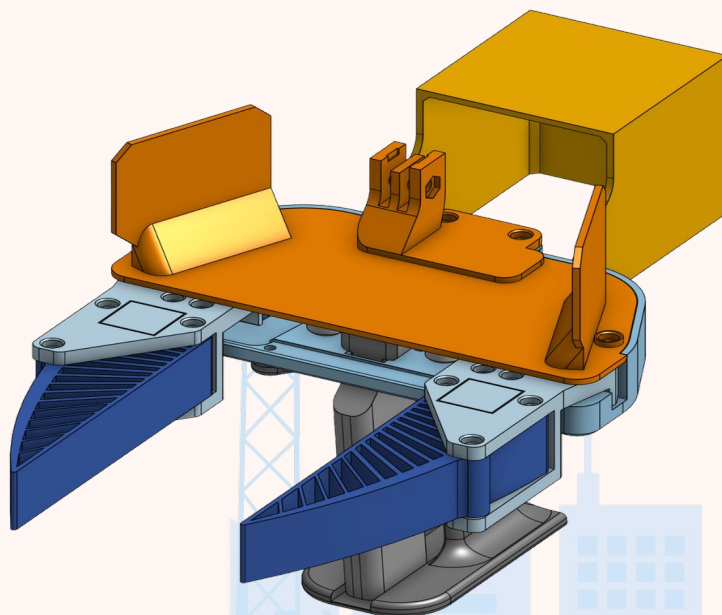


Data Collection Adjustments

Videos 1-100

What we learned

- In this original data collection we made multiple mistakes
- One major issue we had was the time sync of the go pros
- The second issue was our use of 2 mapping videos rather than one
- We also found a film covering the mirrors on UMI Grippers



Videos 100-200

What we learned

- In videos 100 - 200 our largest mistake was our hardware
- once again our cameras slowly moved out of sync of the process of filming
- The UMI device also had one flaw as the QR codes were unfortunately slightly undersized and duplicated on both robots

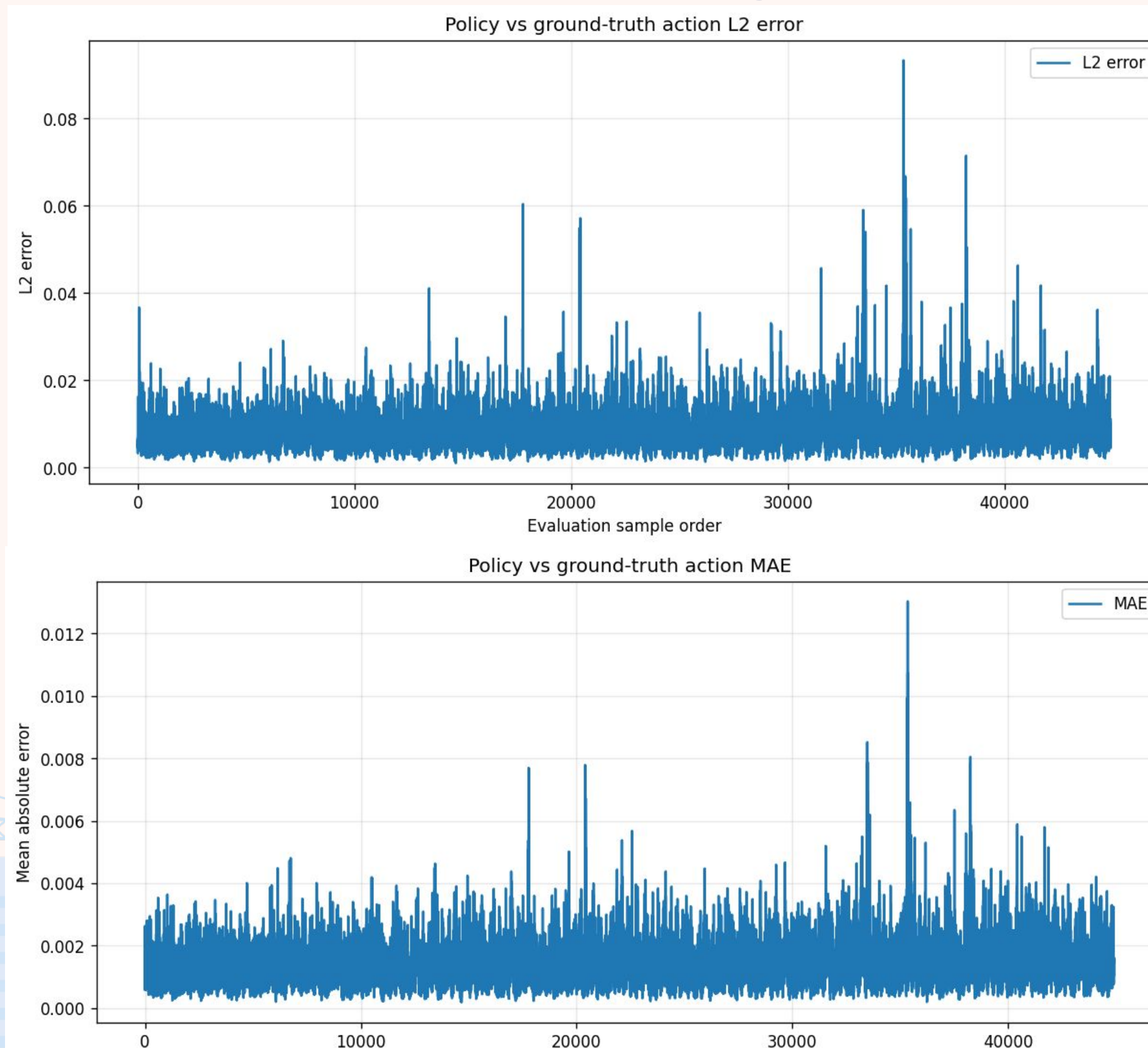
Videos 200-300

Final take aways

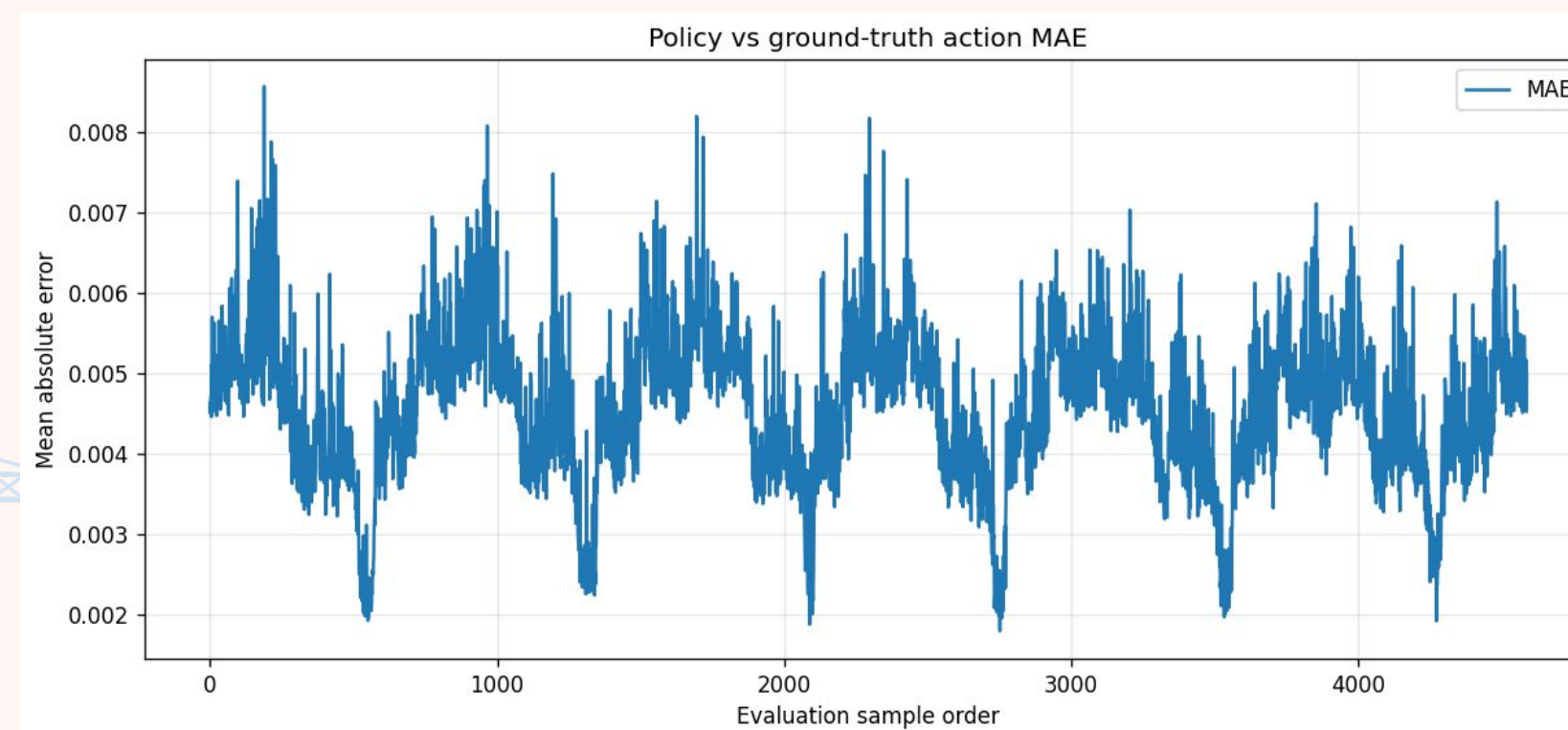
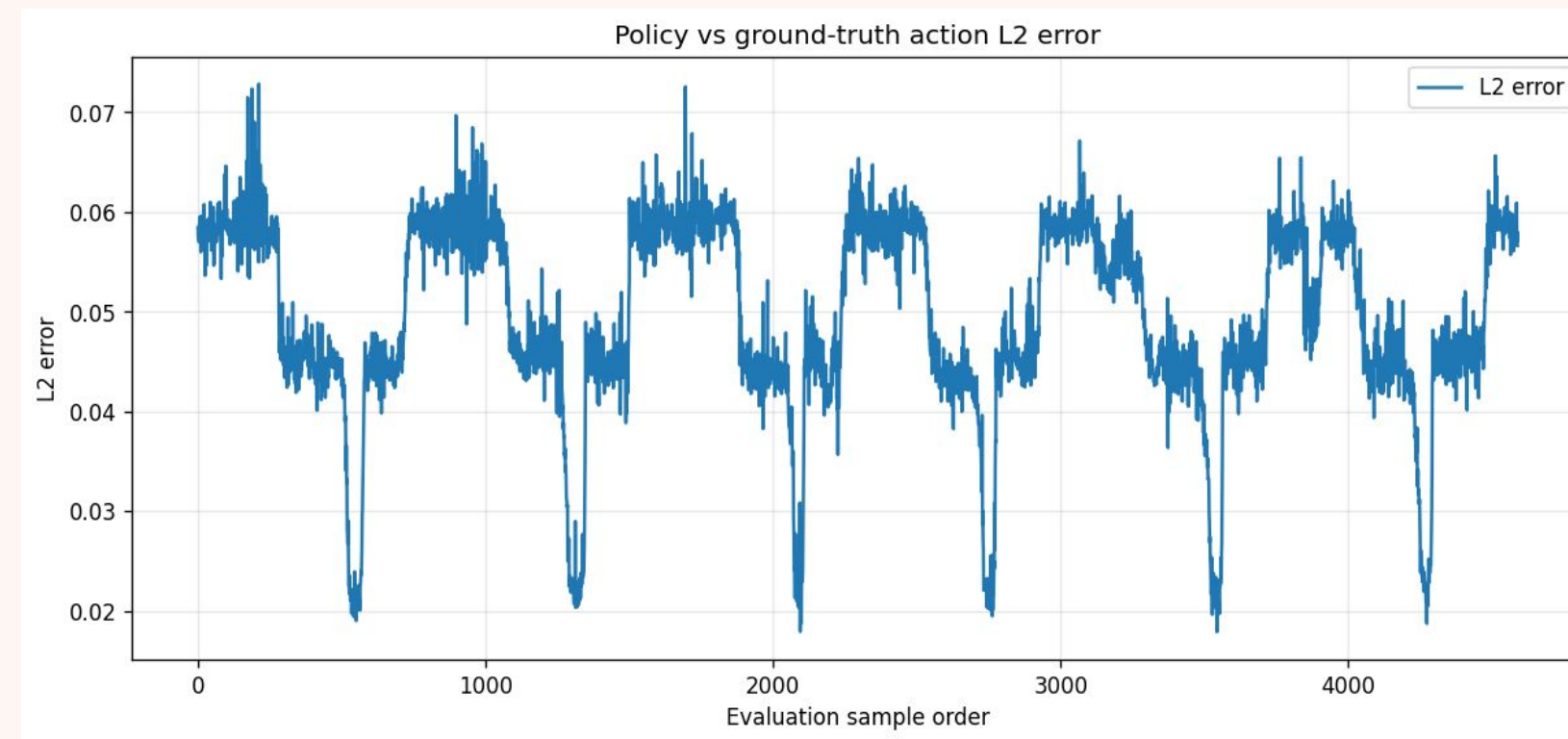
- while filming roughly every 10 demos we implemented rescanning the time sync
- we made the mapping video twice as long
- replaced the QR codes so SLAM can differentiate between arms



Quantitative Analysis: Training



Quantitative Analysis: Testing



Next Steps Forward

- Have been waiting on GoPro Mods necessary for live-streaming video from GoPro to our model
 - Arrived and currently being set-up
- Testing Model Checkpoint on the Physical Hardware (Next Two Days)
 - RX-150 Robot Arms

